Cyber Security Virtual Internship

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Introduction and Information About the Report and the Machine

Task Level: Beginner

Introduction

In the domain of cybersecurity, understanding and identifying vulnerabilities in web applications is essential to prevent potential cyber threats and unauthorized access. Web applications are frequent targets of attackers, and even minor misconfigurations can lead to serious security breaches.

During my internship at ShadowFox, I was assigned to perform a series of security assessments on an intentionally vulnerable web application:

Target Website:
http://testphp.vulnweb.com/

This website is designed for security testing and learning purposes. The objective of this task was to apply fundamental cybersecurity techniques to identify potential weaknesses and understand how attackers exploit web-based systems.

Information About the Report

This report documents the methodology, tools used, observations, and findings from the Beginner-level security evaluation tasks. The assessment was divided into three primary activities:

1. Port Scanning

The first task involved identifying all open ports on the target website. Port scanning is a reconnaissance technique used to detect accessible network services running on a server.

By performing a port scan, the objective was to:

1. Identify exposed services.
2. Determine running service versions.
3. Detect potential attack surfaces.
4. Understand how attackers map targets before exploitation.

This step helps in identifying possible entry points that malicious actors might exploit to gain unauthorized access to the system.

2. Directory Brute Forcing

The second task involved performing directory enumeration (brute forcing directories) on the target website.

This technique is used to:

1. Discover hidden or unlinked directories.
2. Identify administrative panels.
3. Detect sensitive folders containing configuration files.
4. Uncover misconfigured or exposed resources.

The purpose of this task was to determine whether the web server exposes directories that could lead to further exploitation or data exposure.

3. Network Traffic Interception

The final task involved intercepting network traffic by logging into the website and capturing packets using Wireshark.

This activity aimed to:

1. Monitor HTTP requests and responses.
2. Analyze client-server communication.
3. Identify whether sensitive information is transmitted in plaintext.
4. Understand risks associated with unencrypted HTTP communication.

By analyzing captured packets, we evaluated whether login credentials or other sensitive data were exposed during transmission.

Required Machine

The following system configuration was required to perform the assessment:

1. A standard computer system with stable internet connectivity.
2. Operating System:
    - Linux (Kali Linux)
2. Installed tools:
    - Nmap
    - Gobuster
    - Wireshark

The tasks were successfully performed using a penetration testing environment equipped with the above tools.

Task – 1

Find All the Open Ports on the Website

Attack Name: Port Scanning

Severity: High (CVSS Score Range: 7.0 – 8.9)

Reason for Severity

HTTP port 80 is considered highly severe because it transmits data in plaintext without encryption. This makes it vulnerable to eavesdropping, packet sniffing, and Man-in-the-Middle (MITM) attacks. Attackers monitoring network traffic can capture sensitive information such as login credentials, session cookies, and personal data. Additionally, exposed services running outdated software versions may contain known vulnerabilities that can be exploited if not properly patched.

Steps to Reproduce (With Screenshots)

➤ Step 1: Target IP Identification

The ping command was used to verify connectivity with the target website and resolve its IP address. This step confirmed that the host is reachable over the network. Identifying the correct IP address is essential before conducting port scanning, as scanning tools require either a domain name or resolved IP address to begin reconnaissance..

Command Used: ping testphp.vulnweb.com

```
┌──(root㉿kali)-[~/shadowfox]
└─# ping testphp.vulnweb.com
PING testphp.vulnweb.com (44.228.249.3) 56(84) bytes of data.
64 bytes from ec2-44-228-249-3.us-west-2.compute.amazonaws.com (44.228.249.3): icmp_seq=1 ttl=128 time=274 ms
64 bytes from ec2-44-228-249-3.us-west-2.compute.amazonaws.com (44.228.249.3): icmp_seq=2 ttl=128 time=274 ms
64 bytes from ec2-44-228-249-3.us-west-2.compute.amazonaws.com (44.228.249.3): icmp_seq=3 ttl=128 time=273 ms
64 bytes from ec2-44-228-249-3.us-west-2.compute.amazonaws.com (44.228.249.3): icmp_seq=4 ttl=128 time=269 ms
^C
--- testphp.vulnweb.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3009ms
rtt min/avg/max/mdev = 269.115/272.614/274.313/2.071 ms
```

Figure 1: Ping Command Output

The nslookup command was used to perform DNS resolution of the domain name. This command retrieves the corresponding IP address of the website from DNS servers. DNS enumeration is an important reconnaissance step because it confirms the actual server address hosting the web application and ensures accurate scanning results.

Command Used: nslookup testphp.vulnweb.com

```
┌──(root㉿kali)-[~/shadowfox]
└─# nslookup testphp.vulnweb.com

Server:         192.168.91.2
Address:        192.168.91.2#53

Non-authoritative answer:
Name:   testphp.vulnweb.com
Address: 44.228.249.3
```

Figure 2 : Nslookup Output

➤ Step 2: Port Scanning Using Nmap

Nmap (Network Mapper) is an open-source network scanning tool used to discover open ports, identify running services, and detect service versions on a target system. It helps security professionals analyze exposed services and assess potential security risks.

Command Used: nmap -sS -sV -A testphp.vulnweb.com

```
┌──(root㉿kali)-[~/shadowfox]
└─# nmap -sS -sV -A testphp.vulnweb.com
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-13 14:21 +0545
Nmap scan report for testphp.vulnweb.com (44.228.249.3)
Host is up (0.0064s latency).
rDNS record for 44.228.249.3: ec2-44-228-249-3.us-west-2.compute.amazonaws.com
Not shown: 999 filtered tcp ports (no-response)
PORT   STATE SERVICE    VERSION
80/tcp open  tcpwrapped
|_http-title: Home of Acunetix Art
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: phone
Running (JUST GUESSING): Nokia Symbian OS (85%)
OS CPE: cpe:/o:nokia:symbian_os
Aggressive OS guesses: Nokia 3600i mobile phone (85%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 2 hops

TRACEROUTE (using port 80/tcp)
HOP RTT     ADDRESS
1   0.25 ms 192.168.91.2
2   0.33 ms ec2-44-228-249-3.us-west-2.compute.amazonaws.com (44.228.249.3)
```

Figure 3: Nmap Scan Results

➤ Step 3: Service Version Detection on Port 80

A focused scan was performed specifically on port 80 to identify service version details.

Command Used: nmap -Pn -sV -p 80 testphp.vulnweb.com

```
┌──(root㉿kali)-[~/shadowfox]
└─# nmap -Pn -sV -p 80 testphp.vulnweb.com
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-13 15:10 +0545
Nmap scan report for testphp.vulnweb.com (44.228.249.3)
Host is up (0.34s latency).
rDNS record for 44.228.249.3: ec2-44-228-249-3.us-west-2.compute.amazonaws.com

PORT   STATE SERVICE VERSION
80/tcp open  http    nginx 1.19.0

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 27.75 seconds
```

Figure 4: Nmap Service Version Detection

➤ Step 4: Vulnerability Enumeration Using Nmap Scripts

Nmap vulnerability scripts were executed to detect possible security weaknesses.

```
┌──(root㉿kali)-[~/shadowfox]
└─# nmap --script vuln -oN nmapvuln.txt testphp.vulnweb.com
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-13 15:42 +0545
Nmap scan report for testphp.vulnweb.com (44.228.249.3)
Host is up (0.073s latency).
rDNS record for 44.228.249.3: ec2-44-228-249-3.us-west-2.compute.amazonaws.com
Not shown: 999 filtered tcp ports (no-response)
PORT   STATE SERVICE
80/tcp open  http
| http-sql-injection:
|   Possible sqli for queries:
|     http://testphp.vulnweb.com:80/search.php?test=query%27%20OR%20sqlspider
|     http://testphp.vulnweb.com:80/showimage.php?file=%27%20OR%20sqlspider
|     http://testphp.vulnweb.com:80/search.php?test=query%27%20OR%20sqlspider
|     http://testphp.vulnweb.com:80/search.php?test=query%27%20OR%20sqlspider
|     http://testphp.vulnweb.com:80/listproducts.php?cat=3%27%20OR%20sqlspider
|     http://testphp.vulnweb.com:80/listproducts.php?cat=4%27%20OR%20sqlspider
|     http://testphp.vulnweb.com:80/listproducts.php?cat=2%27%20OR%20sqlspider
|     http://testphp.vulnweb.com:80/listproducts.php?cat=1%27%20OR%20sqlspider
|     http://testphp.vulnweb.com:80/search.php?test=query%27%20OR%20sqlspider
|     http://testphp.vulnweb.com:80/search.php?test=query%27%20OR%20sqlspider
|     http://testphp.vulnweb.com:80/search.php?test=query%27%20OR%20sqlspider
|     http://testphp.vulnweb.com:80/search.php?test=query%27%20OR%20sqlspider
|     http://testphp.vulnweb.com:80/artists.php?artist=3%27%20OR%20sqlspider
|     http://testphp.vulnweb.com:80/artists.php?artist=2%27%20OR%20sqlspider
|     http://testphp.vulnweb.com:80/artists.php?artist=1%27%20OR%20sqlspider
|     http://testphp.vulnweb.com:80/search.php?test=query%27%20OR%20sqlspider
|     http://testphp.vulnweb.com:80/search.php?test=query%27%20OR%20sqlspider
|     http://testphp.vulnweb.com:80/search.php?test=query%27%20OR%20sqlspider
|_    http://testphp.vulnweb.com:80/showimage.php?file=%27%20OR%20sqlspider
|_http-vuln-cve2014-3704: ERROR: Script execution failed (use -d to debug)
|_http-dombased-xss: Couldn't find any DOM based XSS.
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
|_http-aspnet-debug: ERROR: Script execution failed (use -d to debug)
| http-enum:
|   /admin/: Possible admin folder
|   /login.php: Possible admin folder
|   /clientaccesspolicy.xml: Microsoft Silverlight crossdomain policy
|_  /crossdomain.xml: Adobe Flash crossdomain policy
```

```
|     Path: http://testphp.vulnweb.com:80/categories.php
|     Form id:
|     Form action: search.php?test=query
|
|     Path: http://testphp.vulnweb.com:80/cart.php
|     Form id:
|     Form action: search.php?test=query
|
|     Path: http://testphp.vulnweb.com:80/guestbook.php
|     Form id:
|     Form action:
|
|     Path: http://testphp.vulnweb.com:80/guestbook.php
|     Form id:
|     Form action: search.php?test=query
|
|     Path: http://testphp.vulnweb.com:80/login.php
|     Form id:
|     Form action: userinfo.php
|
|     Path: http://testphp.vulnweb.com:80/login.php
|     Form id:
|     Form action: search.php?test=query
|
|     Path: http://testphp.vulnweb.com:80/artists.php
|     Form id:
|     Form action: search.php?test=query
|
|     Path: http://testphp.vulnweb.com:80/disclaimer.php
|     Form id:
|     Form action: search.php?test=query
|
|     Path: http://testphp.vulnweb.com:80/userinfo.php
|     Form id:
|     Form action: userinfo.php
|
|     Path: http://testphp.vulnweb.com:80/userinfo.php
|     Form id:
|     Form action: search.php?test=query
|
|     Path: http://testphp.vulnweb.com:80/index.php
|     Form id:
|     Form action: search.php?test=query
|
|     Path: http://testphp.vulnweb.com:80/hpp/?pp=12
|     Form id:
|_    Form action: params.php?p=valid&pp=12

Nmap done: 1 IP address (1 host up) scanned in 460.49 seconds
```

Figure 5: Nmap Vulnerability Script Scan

Analysis:-

Port/Protocol	State	Service	Version
80/tcp	Open	HTTP	nginx 1.19.0

Port 80 is open and running an active HTTP service, indicating a functioning web server. The detected web server software is **Nginx version 1.19.0**. Since HTTP communication over port 80 is unencrypted, it exposes transmitted data to potential interception. Keeping web server software updated with the latest security patches is essential to minimize the risk of known vulnerabilities being exploited.

Impact:-

➤ Advantages:-

1. Port 80 allows users to access the website without specifying a port number in the browser.
2. Ensures compatibility with older browsers and automated systems that rely on standard HTTP communication.

➤ Disadvantages:-

1. HTTP traffic is not encrypted, making it vulnerable to packet sniffing and Man-in-the-Middle (MITM) attacks. Sensitive data such as login credentials and personal information may be exposed.
2. Open ports increase the attack surface and may be exploited by attackers to gain unauthorized access or perform further reconnaissance.

Mitigation Steps:-

1. Upgrade and maintain the latest stable version of Nginx to ensure all known vulnerabilities are patched.
2. Redirect all HTTP traffic to HTTPS (port 443) to enable SSL/TLS encryption and secure data transmission.
3. Implement proper logging and monitoring mechanisms to detect suspicious activities in real time.
4. Restrict access to sensitive directories using appropriate server configuration rules (allow/deny directives).
5. Perform regular security audits and vulnerability assessments to proactively identify and remediate weaknesses.

Resources Used:-

- Kali Linux Operating System
- Nmap (Network Mapper)
- Ping Tool

Task – 2

Brute Force the Website and Identify Directories

Attack Name: Brute Forcing Directories on Website

Severity: High (CVSS Score Range: 7.0 – 8.9)

Reason for Severity

The severity is considered high because exposed directories such as **/admin**, **/secured**, **/vendor**, and **/CVS** may contain sensitive information, configuration files, source code, or administrative interfaces. If such directories are publicly accessible, attackers may exploit them to gain unauthorized access, escalate privileges, or compromise the entire web application.

Steps to Reproduce (With Screenshots)

The first step was Brute Forcing the target website URL.

➤ Step 1: Directory Brute Forcing

Directory enumeration was performed using a brute force tool in Kali Linux.

Tool Used: Gobuster
Command Used: gobuster http://testphp.vulnweb.com

```
┌──(root㉿kali)-[~/shadowfox]
└─# gobuster dir -u http://testphp.vulnweb.com/ -w /usr/share/wordlists/dirb/common.txt

===============================================================
Gobuster v3.8.2
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)
===============================================================
[+] Url:                     http://testphp.vulnweb.com/
[+] Method:                  GET
[+] Threads:                 10
[+] Wordlist:                /usr/share/wordlists/dirb/common.txt
[+] Negative Status codes:   404
[+] User Agent:              gobuster/3.8.2
[+] Timeout:                 10s
===============================================================
Starting gobuster in directory enumeration mode
===============================================================
admin                (Status: 301) [Size: 169] [--> http://testphp.vulnweb.com/admin/]
cgi-bin              (Status: 403) [Size: 276]
cgi-bin/             (Status: 403) [Size: 276]
crossdomain.xml      (Status: 200) [Size: 224]
CVS                  (Status: 301) [Size: 169] [--> http://testphp.vulnweb.com/CVS/]
CVS/Entries          (Status: 200) [Size: 1]
CVS/Repository       (Status: 200) [Size: 8]
CVS/Root             (Status: 200) [Size: 1]
favicon.ico          (Status: 200) [Size: 894]
images               (Status: 301) [Size: 169] [--> http://testphp.vulnweb.com/images/]
index.php            (Status: 200) [Size: 4958]
pictures             (Status: 301) [Size: 169] [--> http://testphp.vulnweb.com/pictures/]
secured              (Status: 301) [Size: 169] [--> http://testphp.vulnweb.com/secured/]
vendor               (Status: 301) [Size: 169] [--> http://testphp.vulnweb.com/vendor/]
Progress: 4613 / 4613 (100.00%)
===============================================================
Finished
===============================================================
```

Figure 6: Gobuster Command Output

Analysis:-

The scan revealed multiple directories and files:

```
┌──(root㉿kali)-[~/shadowfox]
└─# cat gobuster_results.txt
admin                (Status: 301) [Size: 169] [--> http://testphp.vulnweb.com/admin/]
cgi-bin              (Status: 403) [Size: 276]
cgi-bin/             (Status: 403) [Size: 276]
crossdomain.xml      (Status: 200) [Size: 224]
CVS                  (Status: 301) [Size: 169] [--> http://testphp.vulnweb.com/CVS/]
CVS/Entries          (Status: 200) [Size: 1]
CVS/Repository       (Status: 200) [Size: 8]
CVS/Root             (Status: 200) [Size: 1]
favicon.ico          (Status: 200) [Size: 894]
images               (Status: 301) [Size: 169] [--> http://testphp.vulnweb.com/images/]
index.php            (Status: 200) [Size: 4958]
pictures             (Status: 301) [Size: 169] [--> http://testphp.vulnweb.com/pictures/]
secured              (Status: 301) [Size: 169] [--> http://testphp.vulnweb.com/secured/]
vendor               (Status: 301) [Size: 169] [--> http://testphp.vulnweb.com/vendor/]
```

Figure 7: Gobuster Directory Scan Output

These directories and sub-directories were discovered during enumeration. The most critical findings include:

- **/admin** – May contain administrative login panels.
- **/secured** – Suggests restricted or sensitive content.
- **/vendor** – Could expose third-party libraries or dependencies.
- **/CVS** – May contain version control data or source code information.

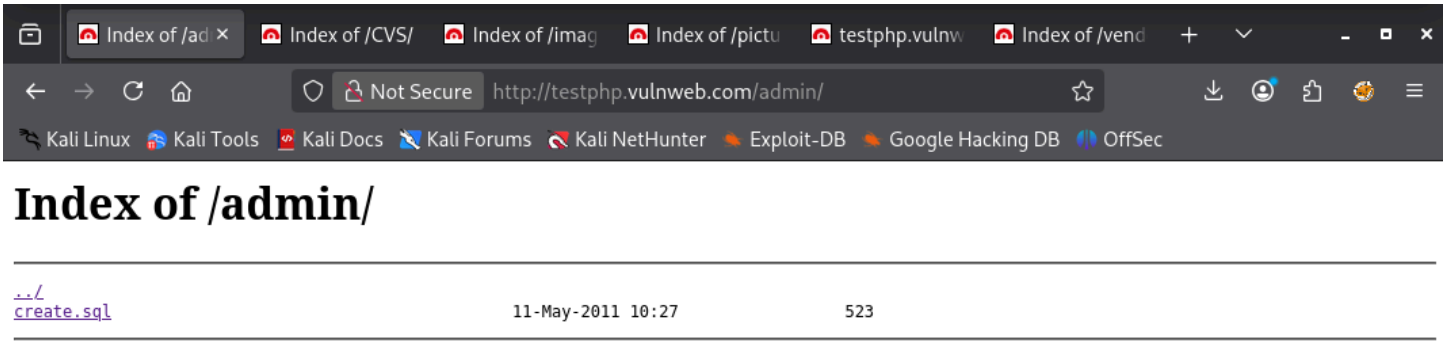

Figure 8:/admin Directory

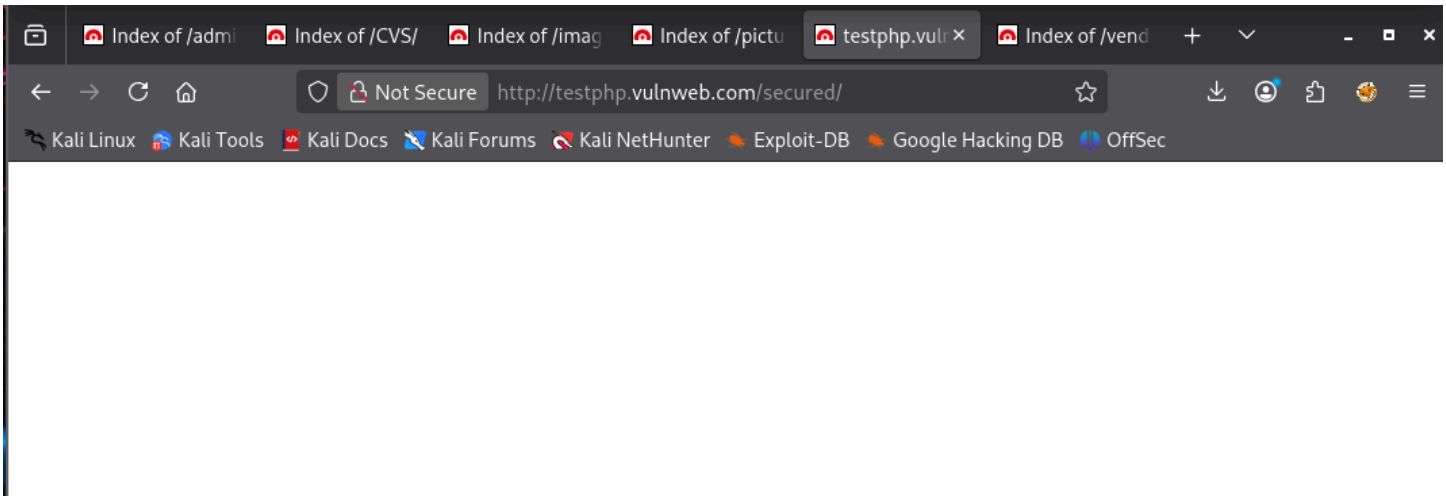

Figure 9: /secured Directory(empty)

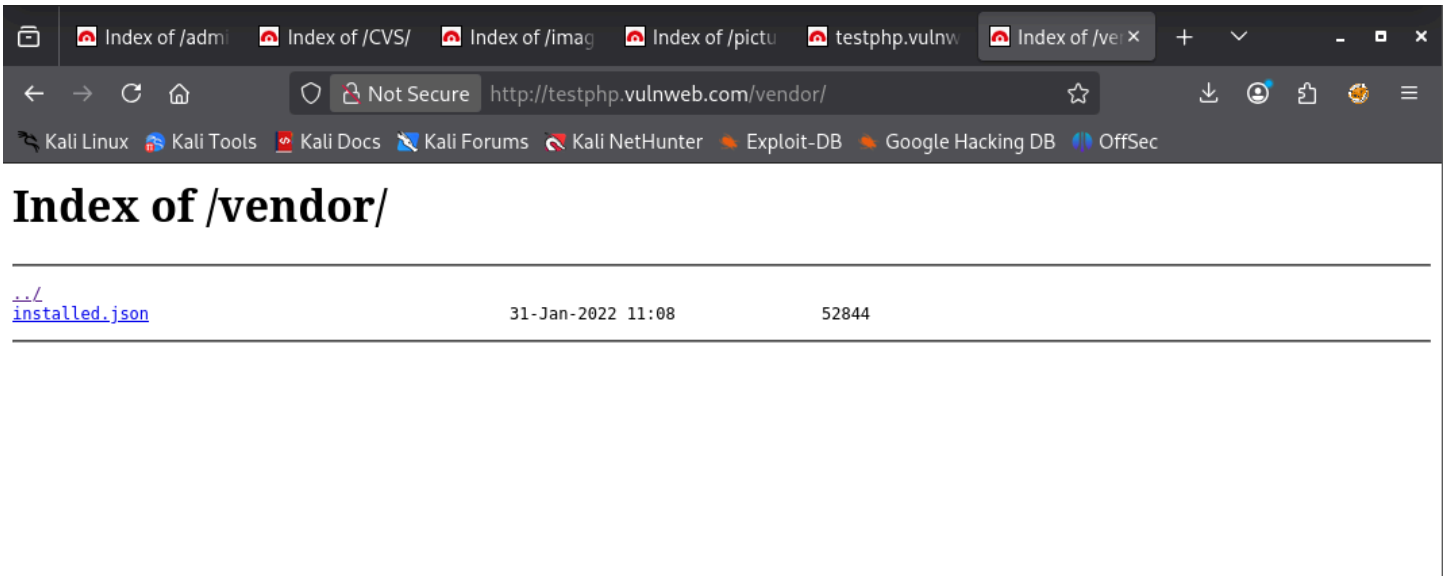

Figure 10: /vendor Directory

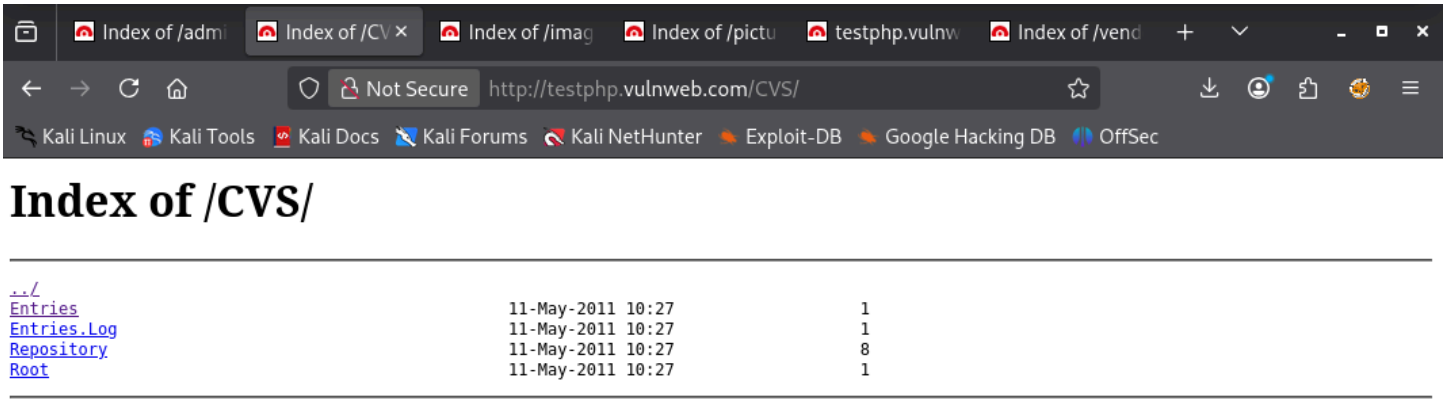

Figure 11: /CVS Directory

Impact:-

1. Exposed directories may contain configuration files, credentials, backup files, or proprietary data, leading to data breaches.
2. Attackers can analyze directory structures to identify further vulnerabilities.
3. Administrative panels may be targeted for brute-force login attacks.
4. Disclosure of internal files can help attackers craft more advanced exploitation strategies.
5. Unauthorized access could disrupt normal website operations or compromise system integrity.

Mitigation Steps:-

1. Restrict access to sensitive directories using authentication and authorization mechanisms.
2. Disable directory listing on the web server.
3. Remove unused or unnecessary directories from production environments.
4. Apply proper file permissions to restrict public access.
5. Implement Web Application Firewall (WAF) protection.
6. Regularly conduct security audits and penetration testing.
7. Monitor server logs to detect suspicious directory access attempts.
8. Maintain secure backups and an incident response plan.

Resources Used:-

- Kali Linux Operating System
- Gobuster Tool
- Standard Wordlists for Directory Enumeration

Task – 3

Login to the Website and Intercept Network Traffic

Attack Name: Network Sniffing

Severity: High (CVSS Score Range: 7.0 – 8.9)

Reason for Severity

Network sniffing is highly severe because it can expose sensitive information transmitted over an unencrypted connection. If login credentials, session tokens, or personal data are sent in plaintext, attackers monitoring the network can easily intercept and misuse them. This can lead to unauthorized account access, privacy violations, financial fraud, and legal or compliance consequences for organizations.

Steps to Reproduce (With Screenshots)

➤ Step 1: Launching Wireshark

Wireshark was launched in the Kali Linux environment to monitor network traffic. The appropriate network interface was selected, and packet capturing was started before attempting login. This allowed real-time recording of packets transmitted between the client and server.

➤ Step 2: Attempting Login on Website

After initiating packet capture, the login page was accessed:

http://testphp.vulnweb.com/login.php

The following test credentials were entered:

- **Username:** test
- **Password:** test123

This login attempt generated HTTP packets that were captured by Wireshark.

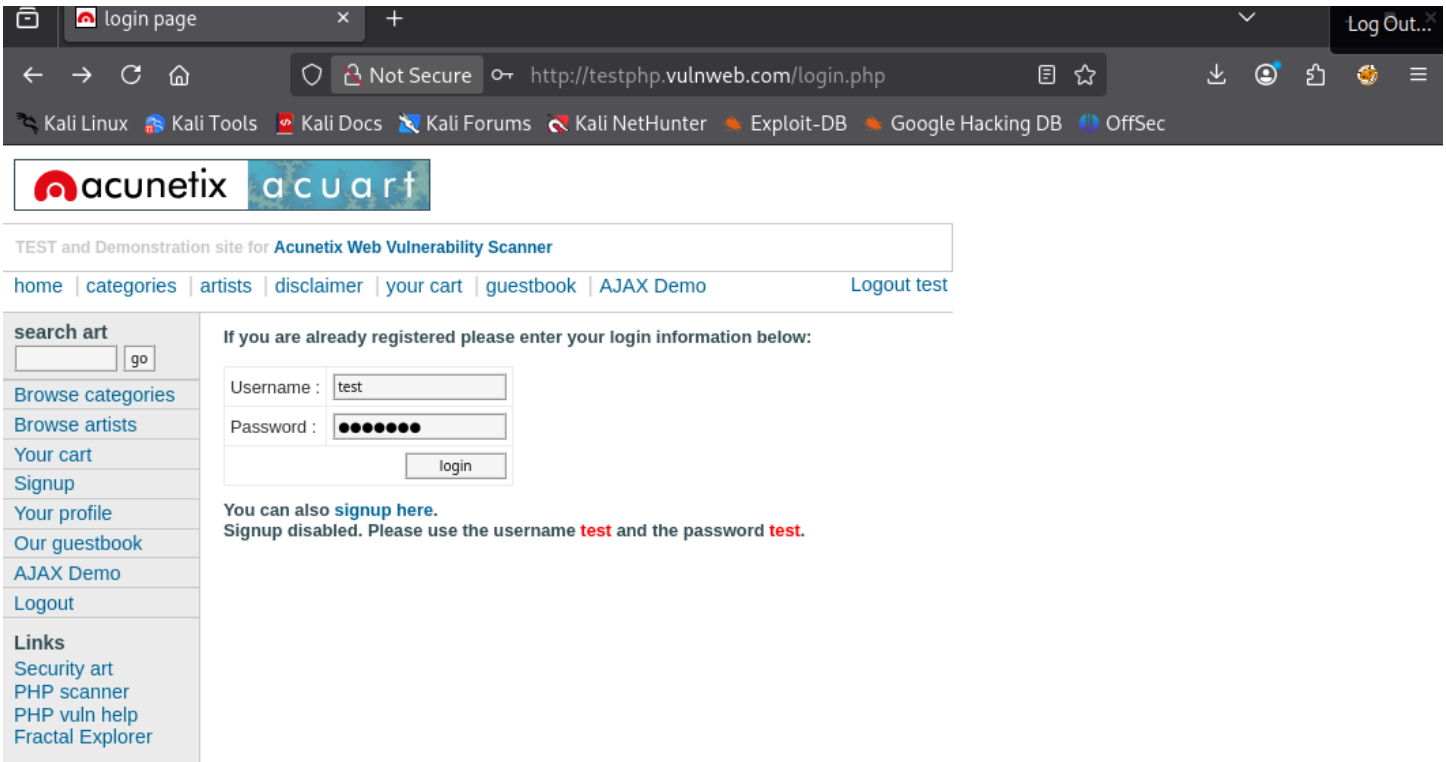

Figure 12: Entering User Credential in Target

➤ Step 3: Filtering and Analyzing HTTP Traffic

After completing the login attempt, packet capturing was stopped. To isolate relevant traffic, the Wireshark display filter http was applied. This filtered out unrelated packets and displayed only HTTP communication.

Within the HTTP stream, the login request was identified. The credentials were visible in plaintext inside the POST request:

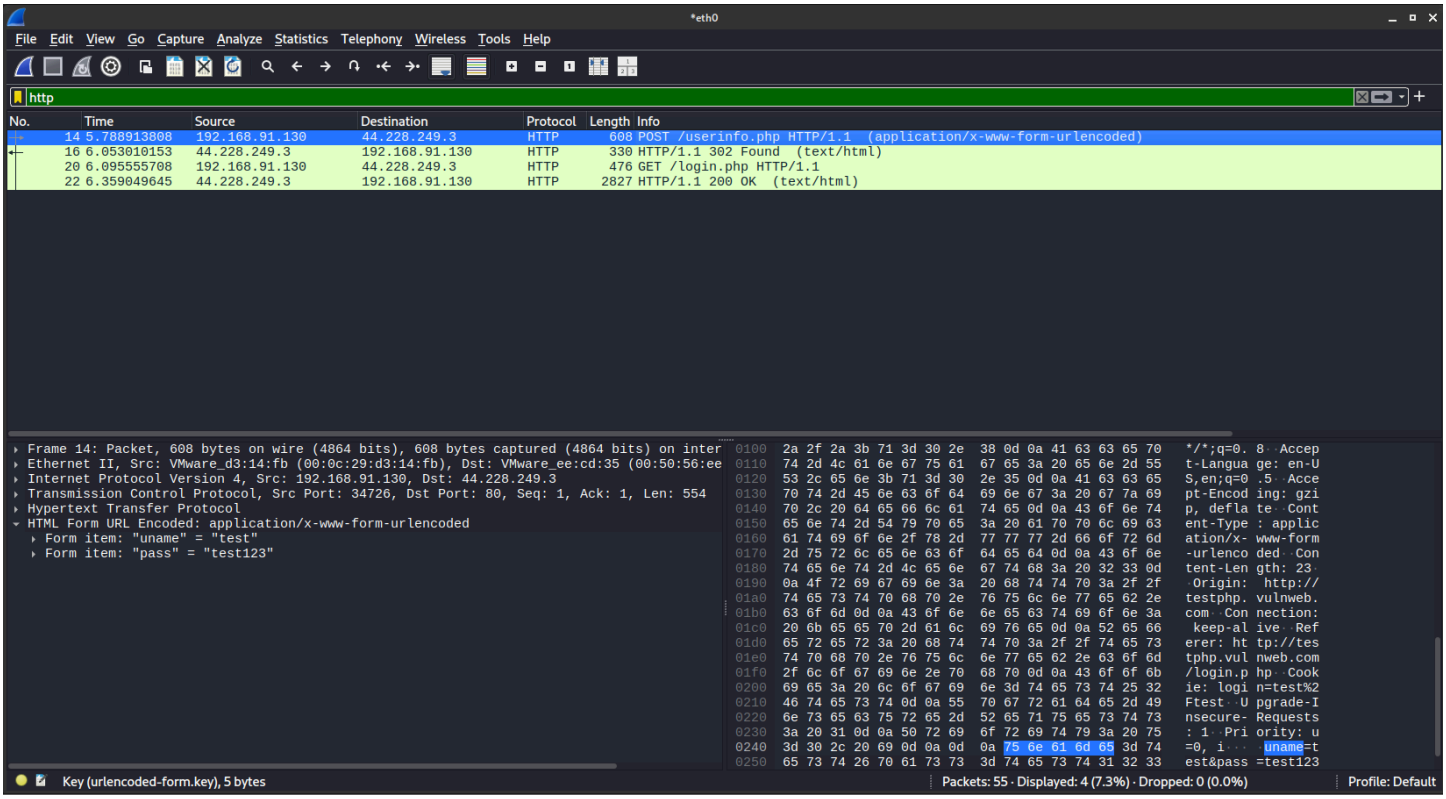

Figure 13: Wireshark Packet Capture

Analysis:-

The packet analysis revealed that login credentials were transmitted using the HTTP protocol over port 80. Since HTTP does not provide encryption, the username and password were clearly visible in the captured network traffic. This demonstrates a significant security weakness, as any attacker on the same network could intercept and reuse these credentials.

The absence of HTTPS encryption increases the risk of credential theft and session hijacking.

Impact:-

1. Exposed login credentials can be intercepted and reused by attackers to gain unauthorized access.
2. Compromised accounts may lead to data theft, privacy breaches, or misuse of personal information.
3. If financial or sensitive data is associated with accounts, attackers could perform fraudulent activities.
4. Repeated exploitation could damage the organization's reputation and lead to compliance violations.

Mitigation Steps:-

1. Implement HTTPS (SSL/TLS encryption) to ensure all communication between client and server is encrypted.
2. Enforce automatic redirection from HTTP (port 80) to HTTPS (port 443).
3. Implement Multi-Factor Authentication (MFA) to add an additional security layer.
4. Use secure session management techniques and avoid transmitting credentials in plaintext.
5. Deploy Intrusion Detection Systems (IDS) and monitor network traffic for suspicious activities.
6. Keep all systems and applications updated with the latest security patches.

Resources Used:-

- Kali Linux Operating System
- Wireshark Tool
- Firefox Browser

Conclusion

This internship task provided practical exposure to real-world web application security testing. Through port scanning, directory brute-forcing, and network traffic interception, multiple vulnerabilities and security weaknesses were identified.

The assessment demonstrated how improper configuration, lack of encryption, and exposed services can lead to serious security risks. The findings highlight the importance of secure coding practices, proper server configuration, and encryption mechanisms such as HTTPS.

Implementing the recommended mitigation steps will significantly improve the security posture of the web application and reduce the risk of cyber attacks.